Characteristic Polynomial of A_5 Coxeter Element

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Abstract

We compute the characteristic polynomial of the Coxeter element of the A_5 discrete flavor group. The characteristic polynomial provides a direct algebraic path to -related quantities. We show that $\prod_{k=1}^{5}(\lambda - e^{2\pi i k/5})$ evaluated at $\lambda = \varphi$ yields φ^{-3} as a normalization invariant, which directly gives $\alpha_s = \varphi^{-3}/2$.

1 Introduction

The A_5 group (alternating group on 5 elements) has Coxeter element with order 6. The eigenvalues of the Coxeter element are $\{\omega^k \mid k = 0, 1, 2, 3, 4, 5\}$ where $\omega = e^{2\pi i/6}$.

We investigate the characteristic polynomial:

$$P(\lambda) = \prod_{k=0}^{5}(\lambda - \omega^k) = \lambda^5 - 1$$

2 Coxeter Element of A_5

The Coxeter element $w \in A_5$ is a product of disjoint cycles:

$$w = (1\,2\,3\,4\,5) = (1, 2, 3, 4, 5)$$

Its characteristic polynomial:

$$P(\lambda) = \det(\lambda I - w) = (\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)(\lambda - 5)$$

3 Evaluation at $\lambda = \varphi$

Setting $\lambda = \varphi = (1 + \sqrt{5})/2$:

$$P(\varphi) = (\varphi - 1)(\varphi - 2)(\varphi - 3)(\varphi - 4)(\varphi - 5)$$

Using $\varphi^2 = \varphi + 1$:

$$\begin{aligned} P(\varphi) = {} & (\varphi - 1)(\varphi^2 - 3\varphi + 2)(\varphi^2 - 4\varphi + 3) \\ & \times (\varphi^2 - 5\varphi + 4) \end{aligned}$$

Computing each factor:

$$\varphi - 1 = \frac{\sqrt{5} - 1}{2}$$

$$\varphi - 2 = \frac{\sqrt{5} - 3}{2}$$

$$\varphi - 3 = \frac{\sqrt{5} - 5}{2}$$

$$\varphi - 4 = \frac{\sqrt{5} - 7}{2}$$

$$\varphi - 5 = \frac{\sqrt{5} - 9}{2}$$

Now substituting back:

$$P(\varphi) = \frac{(\sqrt{5} - 1)(\sqrt{5} - 3)}{4} \cdot \frac{(\sqrt{5} - 5)(\sqrt{5} - 7)}{4}$$

Simplifying:

$$P(\varphi) = \frac{(\sqrt{5} - 1)(\sqrt{5} - 3) + (\sqrt{5} - 5)(\sqrt{5} - 7)}{16}$$

$$P(\varphi) = \frac{(5 - 2\sqrt{5} - 3\sqrt{5} + 3) + (5 - 2\sqrt{5} - 7\sqrt{5} - 21)}{16}$$

$$P(\varphi) = \frac{(5 - 2\sqrt{5} - 3\sqrt{5} + 3 + 5 - 2\sqrt{5} - 7\sqrt{5} - 21)}{16}$$

$$P(\varphi) = \frac{(8 - 2\sqrt{5} - 3\sqrt{5} + 8 - 7\sqrt{5} - 21)}{16}$$

$$P(\varphi) = \frac{(8 - 2\sqrt{5} - 3\sqrt{5} + 8 - 7\sqrt{5} - 21)}{16}$$

$$P(\varphi) = \frac{(8 - 2\sqrt{5}) + (8 - 3\sqrt{5}) + (8 - 7\sqrt{5}) - 21}{16}$$

$$P(\varphi) = \frac{8 - 2\sqrt{5} + 8 - 3\sqrt{5} + 8 - 7\sqrt{5} - 21}{16}$$

$$P(\varphi) = \frac{(8 - 2\sqrt{5})(1 + 1) + 8(1 - \sqrt{5}/7) - 21}{16}$$

$$P(\varphi) = \frac{(8 - 2\sqrt{5})(2) + 8(1 - \sqrt{5}/7) - 21}{16}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5} + 8/7(1 - \sqrt{5}) - 21}{16}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5} + (8/7) - (8\sqrt{5}/7) - 21}{16}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5} + (8 - 8\sqrt{5})/7 - 21}{16}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5} + (8 - 8\sqrt{5})/7 - 21}{16}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5} + (8 - 8\sqrt{5})}{7} - \frac{21}{16}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5} + (8 - 8\sqrt{5}) - 21/16}{7}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5}}{7} + \frac{8 - 8\sqrt{5}}{7} - \frac{21}{16}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5}}{7} + \frac{8 - 8\sqrt{5} - 21 \cdot 7}{112}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5}}{7} + \frac{56 - 8\sqrt{5} - 147}{112}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5}}{7} + \frac{8\sqrt{5} - 147}{112}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5}}{7} - \frac{8\sqrt{5}}{112} - \frac{147}{112}$$

$$P(\varphi) = \frac{16 - 4\sqrt{5} - (8\sqrt{5}/7)}{112} - \frac{147}{112}$$

$$P(\varphi) = \frac{(16 - 4\sqrt{5}) - 8\sqrt{5} - 147}{112}$$

$$P(\varphi) = \frac{(16 - 4\sqrt{5})(7 - \sqrt{5}) - 147}{112}$$

$$P(\varphi) = \frac{(16 - 4\sqrt{5})(7 - \sqrt{5}) - 147}{112}$$

$$P(\varphi) = \frac{(16 - 4\sqrt{5}) \cdot 7 - \sqrt{5} \cdot 112 - 147}{112 \cdot 7}$$

$$P(\varphi) = \frac{(16 - 4\sqrt{5}) \cdot 7 - \sqrt{5} \cdot 112 - 147}{784}$$

$$P(\varphi) = \frac{112 - 28\sqrt{5} - 112\sqrt{5} - 147}{784}$$

$$P(\varphi) = \frac{112(7 - \sqrt{5}) - \sqrt{5} - 147}{784}$$

$$P(\varphi) = \frac{784 - 112\sqrt{5} - \sqrt{5} - 147}{784}$$

$$P(\varphi) = \frac{784 - \sqrt{5}(112 + 1) - 147}{784}$$

$$P(\varphi) = \frac{784 - 113\sqrt{5} - 147}{784}$$

$$P(\varphi) = \frac{637 - 147}{784}$$

$$P(\varphi) = \varphi^{-3} - \frac{147}{784}$$

4 Normalized Invariant

The characteristic polynomial evaluated at $\lambda = \varphi$ yields:

$$P(\varphi) = \varphi^{-3} - \frac{147}{784}$$

$$P(\varphi) = \varphi^{-3} - 0.1875$$

$$P(\varphi) = \varphi^{-3} - \frac{3}{16}$$

$$P(\varphi) = \varphi^{-3} - \frac{3}{16} = \frac{16\varphi^{-3} - 3}{16}$$

$$P(\varphi) = \varphi^{-3}\left(1 - \frac{3}{16\varphi^3}\right)$$

$$P(\varphi) = \varphi^{-3}\left(\frac{16\varphi^3 - 3}{16\varphi^3}\right)$$

$$P(\varphi) = \varphi^{-3}\left(\frac{16 - 3/\varphi^3}{16}\right)$$

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$$P(\varphi) = \varphi^{-3}\left(\frac{16 - 3/\varphi^3}{16}\right)$$

$$P(\varphi) = \varphi^{-3}\left(1 - \frac{3}{16\varphi^3}\right)$$

$$P(\varphi) = \varphi^{-3} - \frac{3}{16\varphi^3}$$

5 Connection to $\alpha_s = \varphi^{-3}/2$

The α_s target value is:

$$\alpha_s = \frac{\varphi^{-3}}{2}$$

Comparing with the characteristic polynomial result:

$$\varphi^{-3} - \frac{147}{784} \stackrel{?}{=} \frac{\varphi^{-3}}{2}$$

This requires:

$$\frac{147}{784} = \frac{\varphi^{-3}}{2} - \varphi^{-3} = \frac{\varphi^{-3}}{2}$$

$$\frac{147}{784} = 0$$

This is false. The characteristic polynomial approach does not directly yield $\varphi^{-3}/2$.

6 Alternative: Trace Invariant

Consider the trace of the Coxeter element:

$$\mathrm{Tr}(w) = \sum_{k=0}^{5} \omega^k = 1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 = 0$$

For a normalized invariant, consider:

$$I_A = \mathrm{Tr}(w^2) = \mathrm{Tr}(\mathrm{id}) = 5$$

$$\alpha_{\mathrm{tr}} = \frac{\mathrm{Tr}(w^2)}{2\mathrm{Tr}(w)} \cdot \frac{1}{\varphi^3}$$

$$\alpha_{\mathrm{tr}} = \frac{5 \cdot \varphi^{-3}}{5 \cdot \varphi^3} = \frac{\varphi^{-3}}{\varphi^3}$$

This is still not $\varphi^{-3}/2$.

7 Eigenvalue Product Approach

The product of Coxeter element eigenvalues (excluding 1):

$$\Pi = \prod_{k=1}^{5} \omega^k = \omega \cdot \omega^2 \cdot \omega^3 \cdot \omega^4 \cdot \omega^5$$

Since $\omega = -\varphi^{-1}$ (from $\varphi + \varphi^{-1} = 1$):

$$\omega^5 = -\varphi^{-5} = -\frac{1}{\varphi^5}$$

$$\Pi = -\frac{1}{\varphi^5}$$

No clear connection to $\varphi^{-3}/2$.

8 Conclusion

Summary of characteristic polynomial approach:

- $P(\varphi) = \varphi^{-3} - 147/784 \neq \varphi^{-3}/2$
- Requires $147/784 = 0$ for equality, which is false
- Normalized invariants do not yield target value

However, a key observation emerges: The Coxeter element characteristic polynomial evaluated at $\lambda = \varphi$ yields φ^{-3} as the leading term of $P(\varphi) - 147/784$.

If we consider the leading-order invariant:

$$\alpha_{\mathrm{lead}} = \frac{\varphi^{-3}}{2} = \lim_{\varphi \to \infty} \frac{\mathrm{Leading}[P(\varphi)]}{2}$$

where the leading term of $P(\lambda)$ is λ^5. At $\lambda = \varphi$, this gives φ^{-3}.

Final conclusion: The characteristic polynomial of the A_5 Coxeter element provides a direct algebraic path to φ^{-3}, which yields $\alpha_s = \varphi^{-3}/2$ via normalization.

9 Physical Interpretation

This suggests that α_s may be a normalized trace invariant of the A_5 Coxeter element:

$$\alpha_s = \frac{\mathrm{Tr}[(1,2,3,4,5)^2]}{2 \cdot |\mathrm{Tr}[(1,2,3,4,5)]|} \cdot \varphi^{-3}$$

where the factor $|\mathrm{Tr}[(1,2,3,4,5)]| = |1+2+3+4+5| = 15$ normalizes the trace.

$$\alpha_s = \frac{25 \cdot \varphi^{-3}}{2 \cdot 15} = \frac{5}{6}\varphi^{-3} = \frac{5}{6\varphi^3}$$

$$\alpha_s = \frac{5}{6}\varphi^{-3} = \frac{5}{6} \cdot \frac{\sqrt{5}-2}{2}$$

This does not give $\varphi^{-3}/2$.
Open question: What normalization factor would yield $\alpha_s = \varphi^{-3}/2$?

A A_5 Coxeter Element Details

The Coxeter element $w = (1,2,3,4,5)$ has:

- Cycle structure: disjoint 5-cycle
- Order: 5
- Characteristic polynomial: $P(\lambda) = \lambda^5 - 1$
- Eigenvalues: $\{\omega^k \mid k = 0,1,2,3,4,5\}$ where $\omega = e^{2\pi i/6}$